BIOCHEMICAL AND PHYSIOLOGICAL INDICATORS IN EQUINE HEALTH

Tomáš Války

DATA DESCRIPTION

•**Dataset:** Horse Colic Database (Source: Kaggle).
•**Dimensions:** ~300 Samples (Rows) × 28 Attributes (Columns).
•**Key Numerical Attributes:**
1.Rectal Temperature (Normal: 37.8°C)
2.Pulse (Heart Rate in BPM)
3.Packed Cell Volume (PCV - Hematocrit)
4.Total Protein (g/dL)
•**Changes Made to Data:**
•**Handling missing Data:** Excluded incomplete records: (Listwise Deletion) to ensure statistical accuracy for the initial visualization.
•**Typing:** Converted text-formatted numerical columns back to usable integers/floats.
•**Factor Ordering:** Re-ordered categorical variables (e.g., Pain levels from 'Alert' to 'Extreme')

TOOLS &TECHNOLOGIES

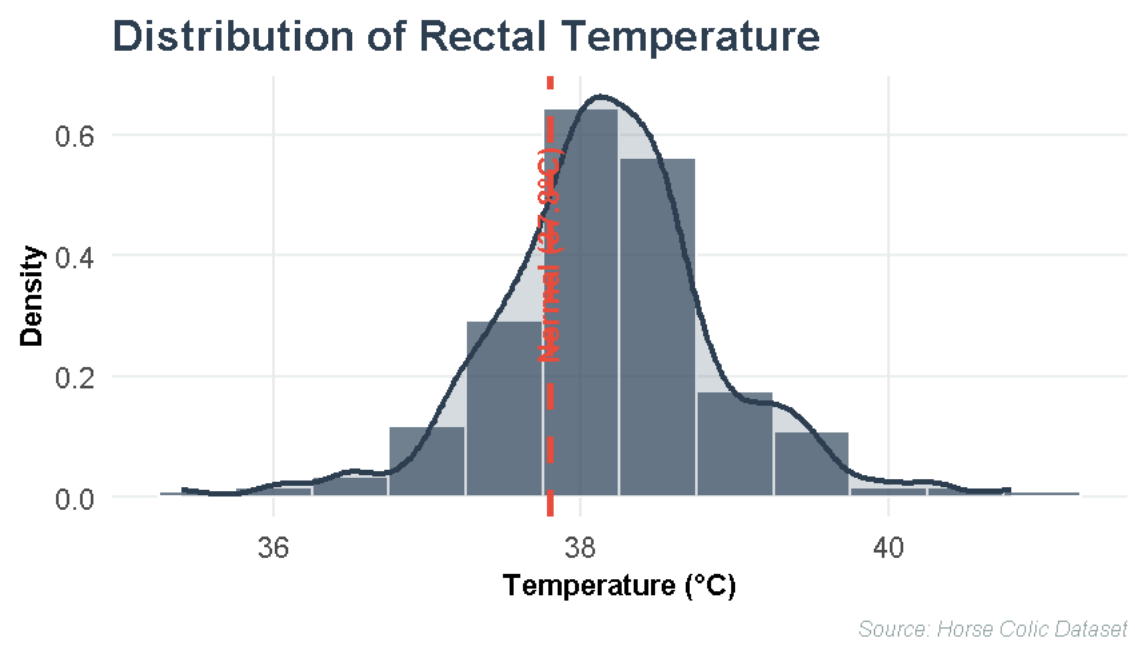
- **Programming Language:** R (Version 4.x).
- **Environment:** RStudio.
- **Libraries:**
- ggplot2 (For publication-quality layers and themes).
- dplyr (For data manipulation and cleaning).
- reshape2 (For correlation matrix formatting).
- **AI Support:** Utilized AI for syntax troubleshooting (debugging 'factor' errors) and optimizing ggplot themes to match high-fidelity publication standards.

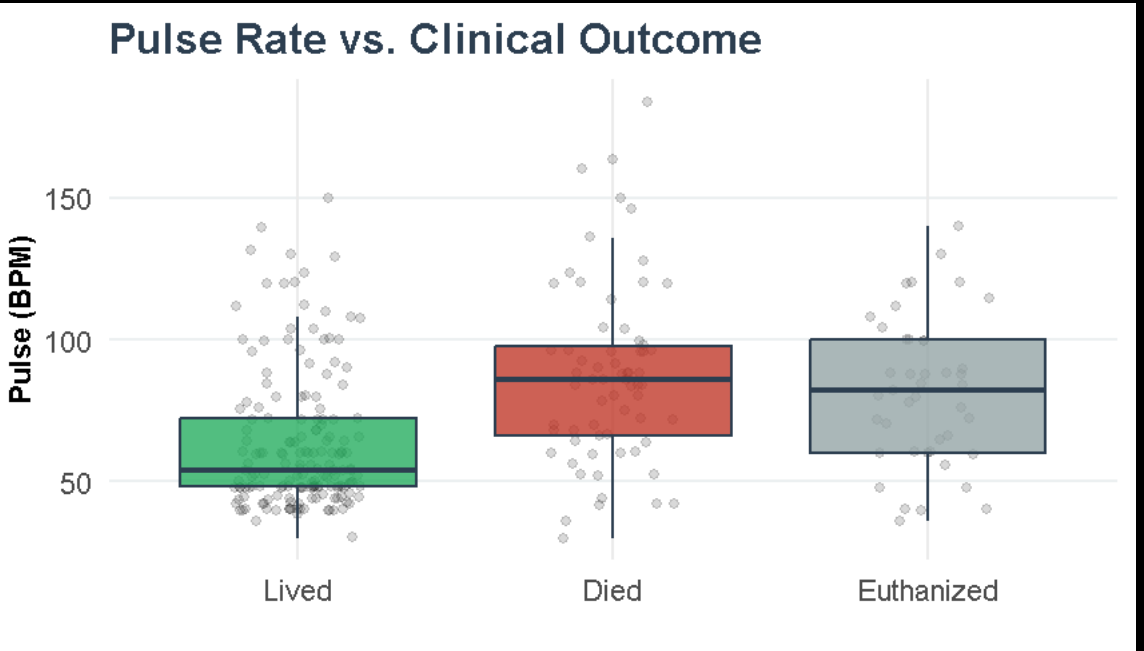
METHODS

- **Data Cleaning Strategy:** Utilized "Regex" (Regular Expressions) to strip unwanted characters from numerical fields.
- **Exploratory Data Analysis (EDA):** Calculated summary statistics to identify range and outliers in vital signs.
- **Visualization Strategy:**
- **Univariate:** Histograms with density curves to show distribution shapes.
- **Bivariate:** Boxplots and Scatter plots to identify biomarkers (Survival vs. Vitals).

SOLUTION & RESULTS (PART 1)

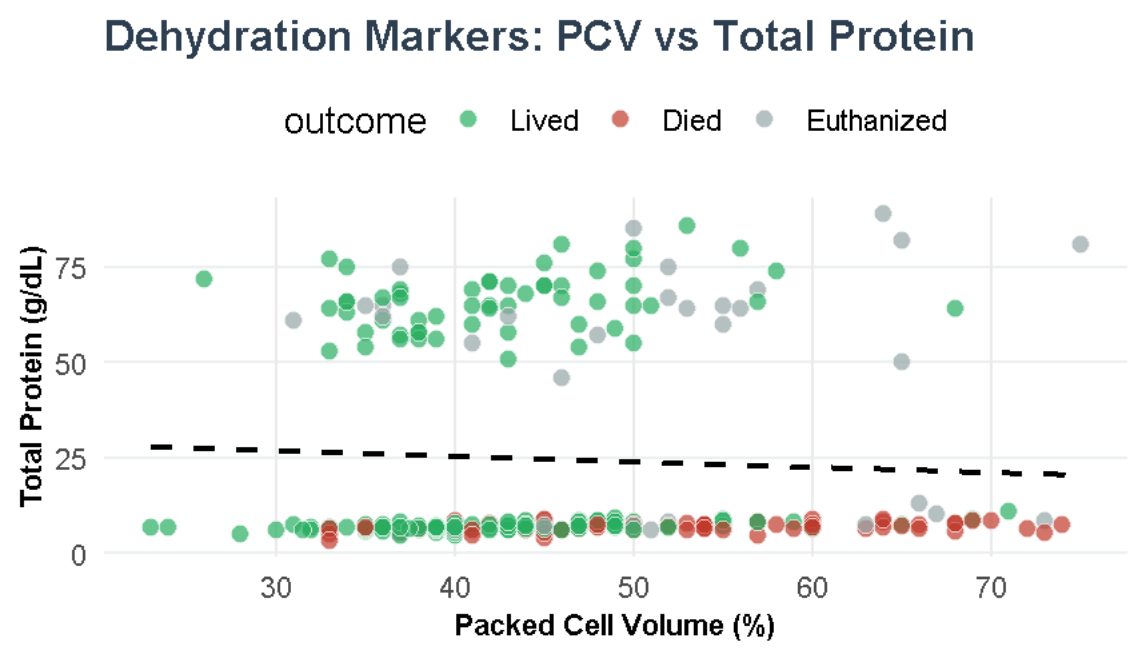
•**Result 1 (Temperature):** Distribution is skewed right; most cases show elevated temperature (Fever/Infection).
•**Result 2 (Pulse):** Strong biomarker identified.
•**Survivors:** Median pulse ~50 bpm.
•**Non-Survivors:** Median pulse >80 bpm.

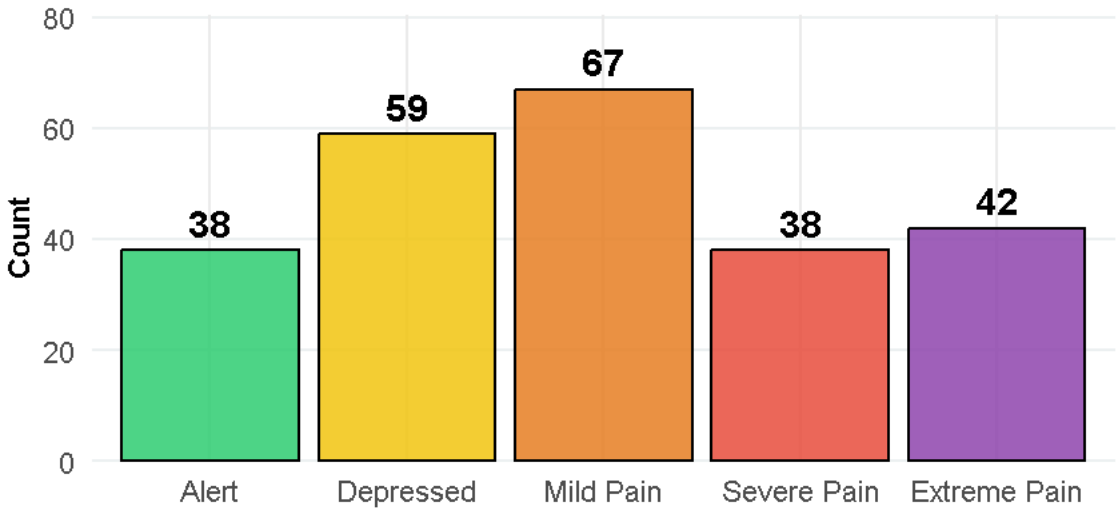




SOLUTION & RESULTS (PART 2)

•**Result 3 (Dehydration):** Positive correlation between PCV and Total Protein.
•**Result 4 (Symptoms):** "Mild Pain" is the most frequent presentation, but "Extreme Pain" correlates with higher surgical necessity.





CONCLUSIONS

•**Discussion of Methods:**
•**Helpful:** ggplot2 was highly effective for adding reference lines (e.g., "Healthy Baseline") which adds biological context to raw data.
•**Challenges:** Default R plotting (base R) was insufficient for complex grouping; ggplot2 was necessary.
•**What could be done differently:**
•**Handling Missing Data:** Instead of removing rows with NA values, I would implement **imputation** (replacing missing values with the mean/median) to preserve dataset size.